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COMPUTER ENGINEERING LAWS & PROFESSIONAL ETHICS

DOCUMENTATION

OPERATING SYSTEMS AND COMPUTER NETWORKS

GROUP 4

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Operating Systems

Operating Systems are system software that act as an intermediary between computer hardware and users or application programs. They manage hardware resources and provide essential services that enable efficient and secure system operation.

The primary purposes of an **Operating System** are to enable applications (softwares) to interact with a computer's hardware and to manage a system's hardware and software resources.

a. Resource Management

Resource management is one of the core responsibilities of an Operating System. It ensures that hardware components such as the CPU, memory, storage, and input or output devices are used efficiently, fairly, and securely. Without proper management, multiple programs running at the same time would conflict with each other, causing crashes, slow performance, or data loss.

The OS is responsible for allocating and controlling system resources to ensure optimal performance:

- **CPU Management**
  CPU management involves deciding which process gets access to the processor and for how long. Since only one process can use a CPU core at a time, the Operating System uses scheduling algorithms to distribute CPU time.

  The OS Performs the following tasks:
  - Creates and deletes processes
  - Schedules processes using algorithms such as First Come First Served (FCFS), Shortest Job First, Priority Scheduling, and Round Robin.
  - Performs context switching
  - Handles multitasking.

  Example:
  In Microsoft Windows, when you open a browser, a music player, and a code editor at the same time, the OS rapidly switches between these processes so they appear to run simultaneously.

Why is it important?
- Prevents CPU starvation
- Improves system responsiveness
- Enables multitasking

- **Memory Management**
  Memory management controls and coordinates computer memory, assigning portions to programs while preventing conflicts. The OS ensures that each process has enough memory to execute while protecting other processes from unauthorized access.

  The OS performs:
  - Allocation and deallocation of RAM
  - Virtual memory management
  - Paging and segmentation
  - Memory protection

  Example:
  If your laptop has 8GB RAM and you open multiple applications, the OS uses virtual memory to move inactive data to disk storage, allowing active programs to continue running.

  Why is it important:
  - Prevents memory conflicts
  - Improves system stability
  - Allows large applications to run even with limited RAM

- **Storage Management (File System Management)**
  Storage management handles data organization, file systems, and disk space allocation. The OS ensures that data is stored efficiently and can be retrieved quickly.

  The OS performs:
  - File creation and deletion
  - Directory management
  - Disk space allocation
  - Access control and permissions

Example:
When you save a document, the OS determines where on the disk it should be stored and records its location in the file system.
Why is it important:

- Organizes data systematically
- Prevents data corruption
- Ensures secure file access

- **Device Management**
  Device management controls communication between the computer and hardware devices such as keyboards, printers, monitors, and storage drives.

  The OS performs:
  - Device driver management
  - Buffering and caching
  - Interrupt handling
  - I/O scheduling

  A device driver acts as a translator between the OS and hardware devices.

  Example:
  When you connect a printer, Microsoft Windows automatically installs a driver so the system can communicate with the printer.

  Why is it important:
  - Enables hardware compatibility
  - Improves I/O performance
  - Ensures efficient communication between software and hardware.

In summary, the resource management allows the operating system to allocate CPU time fairly among processes, manage memory efficiently using virtual memory techniques, organize and secure data using file systems, and control hardware devices through drivers and I/O systems. Through proper resource management, the OS ensures system efficiency, reliability, multitasking capability, and security.

b. *Process Scheduling and Multitasking*

Process scheduling is a core function of the Operating System that enables multitasking. Multitasking allows multiple programs to appear as if they are running at the same time, even on a single CPU. This is possible because the OS rapidly switches the CPU between active processes.

The OS supports multitasking, allowing multiple programs to run seemingly at the same time:

- Uses **scheduling algorithms**
  The Operating System uses scheduling algorithms to determine which process runs next. These algorithms aim to balance fairness, efficiency, and responsiveness. Common scheduling methods include:
  - **First Come First Served (FCFS)**
    In First Come, First Served scheduling, the process that arrives first in the ready queue is the first one to be assigned to the CPU. If two processes arrive at the same time, the one with the smaller process ID is executed first. This scheduling method follows a strict first come, first served basis and is non preemptive in nature, meaning that once a process starts executing, it continues until it finishes or voluntarily releases the CPU. Its implementation is based on the FIFO queue structure, which makes it simple and easy to understand and implement.

    Advantages:
    - It is simple and easy to understand.
    - It can be easily implemented using queue data structure.
    - It does not lead to starvation.

    Disadvantages:
    - It does not consider the priority or burst time of the processes.
    - It suffers from convoy effect i.e. processes with higher burst time arrived before the processes with smaller burst time.

  - **Shortest Job First (SJF)**

In Shortest Job First, the process with the shortest CPU burst time is executed first. If two processes have the same burst time, the First Come, First Served method is used to decide which one runs first. This scheduling approach can be either non preemptive or preemptive. It is considered one of the best methods for minimizing average waiting time, especially in batch systems where the required CPU time of each process is known beforehand. However, it is difficult to apply in interactive systems because the exact burst time of processes is usually not known in advance. For SJF to work properly, the processor must know how long each process will take. The preemptive version of SJF is called Shortest Remaining Time First, where the process with the smallest remaining execution time is always selected.

Advantages:

- SRTF is optimal and guarantees the minimum average waiting time.
- It provides a standard for other algorithms since no other algorithm performs better than it.

Disadvantages:

- It can not be implemented practically since the burst time of the processes can not be known in advance.
- It leads to starvation for processes with larger burst time.
- Priorities can not be set for the processes.
- Processes with larger burst time have poor response time.

- **Priority Scheduling**

  In Priority Scheduling, the CPU is assigned to the process with the highest priority among all available processes. If two or more processes have the same priority, the tie is resolved using First Come, First Served scheduling. This algorithm can operate in either preemptive or non preemptive mode. In preemptive mode, the process with the highest priority will always have zero waiting time because it can immediately take control of the CPU when it arrives. In non preemptive mode, however, the highest priority process may still have to wait if another process is currently executing. Priority

scheduling behaves the same in both modes when all processes arrive at the same time and become available simultaneously.

Advantages:

- It considers the priority of the processes and allows the important processes to run first.
- Priority scheduling in pre-emptive mode is best suited for real time operating systems.

Disadvantages:

- Processes with lesser priority may starve for CPU.
- There is no idea of response time and waiting time.

- **Round Robin**

In Round Robin Scheduling, the CPU is assigned to processes in the order they arrive, following the First Come, First Served principle, but only for a fixed amount of time known as the time quantum or time slice. Once a process uses up its time quantum, it is preempted and placed back into the ready queue, and the CPU is then given to the next process in line. Because processes are regularly interrupted after each time slice, Round Robin is always preemptive in nature.

Advantages:

- It gives the best performance in terms of average response time.
- It is best suited for time sharing system, client server architecture and interactive system

Disadvantages:

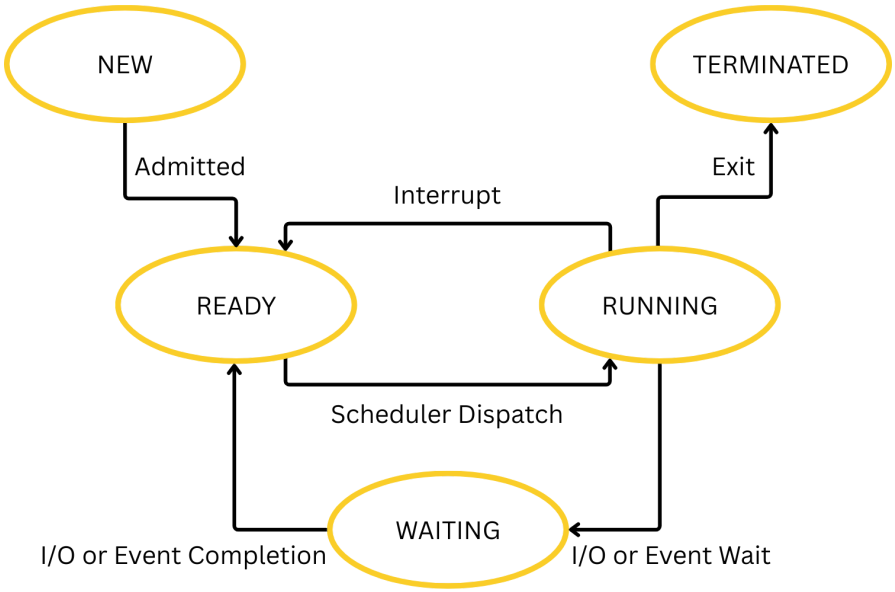
- It leads to starvation for processes with larger burst time as they have to repeat the cycle many times.
- Its performance heavily depends on the time quantum.
- Priorities can not be set for the processes.

When the time quantum is small, the number of context switches increases, response time improves, and the chances of starvation decrease. On the other hand, when the time quantum is large, the

number of context switches decreases, response time becomes slower, and the chances of starvation increase. As the time quantum continues to increase, Round Robin gradually behaves like First Come, First Served scheduling, and if the time quantum approaches infinity, it effectively becomes FCFS. Therefore, the time quantum should be selected carefully so that it is neither too large nor too small in order to balance response time and context switching overhead.

- Manages **process states**

Process State



A process goes through different states from the time it is created until it finishes execution. These states describe the current condition of the process and help the operating system manage CPU scheduling and resource allocation properly.

Each process may be in one of the following states:

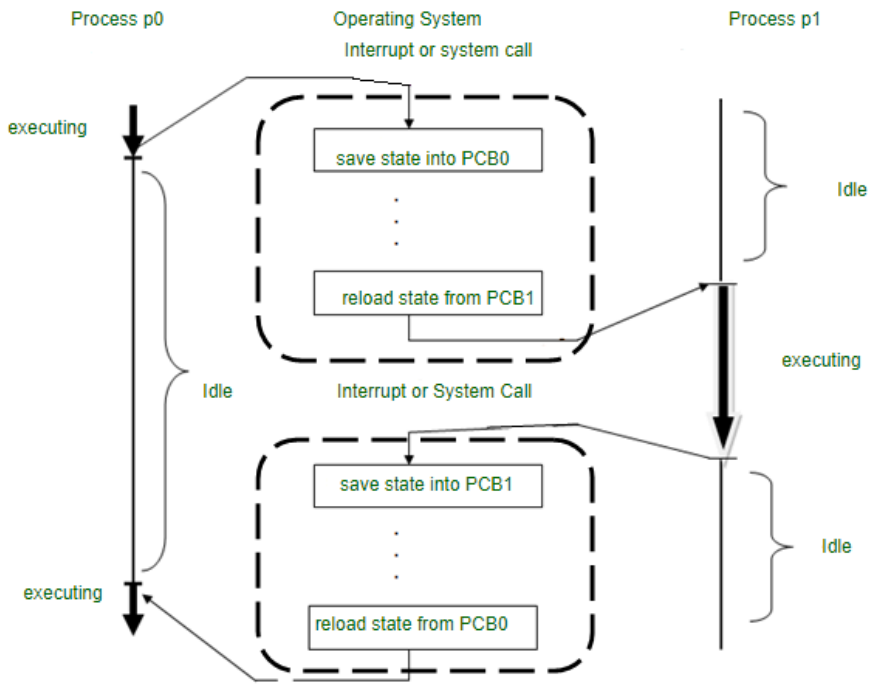
- NEW - the process is being created.

- RUNNING - instructions are being executed.
- WAITING - the process is waiting for some event to occur (such as an I/O completion or reception of a signal).
- READY - the process is waiting to be assigned to a processor.
- TERMINATED - the process has finished execution.

A process begins in the NEW state when it is being created. At this state, the operating system prepares the necessary data structures, such as the Process Control Block. Once the process is ready to run and waiting for CPU time, it enters the READY state. When the scheduler selects it and assigns the CPU, the process moves to the RUNNING state, where its instructions are actively executed.

If the running process needs to wait for an event such as input or output completion, it transitions to the WAITING state. It remains there until the required event occurs. After the event is completed, the process returns to the READY state to wait for CPU time again. Finally, once the process finishes execution, it enters the TERMINATED state, where the operating system releases its resources.

- Enables **context switching** to quickly switch the CPU between processes.



Context switching is a fundamental operation in an operating system where the CPU changes from executing one process to executing another. Since a processor can handle only one instruction stream at a time, the operating system must frequently switch between tasks to support multitasking. This switch usually happens because of interrupts, which are internal or external signals that require immediate CPU attention, or when a higher priority process becomes ready to run and must take control of the CPU. These operations are common in general purpose systems where many applications run simultaneously.

This ensures responsiveness and fairness among users and applications.

c. *User-System Interface*

The OS provides interfaces through which users interact with the computer:

- **Command-Line Interface (CLI)**
  A Command Line Interface, commonly known as CLI, is a text based user interface that allows users to interact with the operating system by typing commands. Instead of clicking icons or buttons, users enter specific instructions through a keyboard. Examples include the Command Prompt in Windows Command Prompt, the Terminal in Linux distributions, and the Terminal application in macOS.

  Characteristics
  A CLI is text based and relies entirely on typed commands. Users must know the correct syntax and structure of commands to execute tasks properly. It provides direct communication between the user and the operating system, often through a shell program. CLIs are typically lightweight and consume very little system memory compared to graphical interfaces. They also support scripting, which allows users to automate repetitive tasks by writing command sequences in script files. Additionally, CLIs are highly precise, meaning even small typing errors can cause commands to fail.

  Advantages
  One major advantage of a CLI is speed and efficiency for experienced users. Tasks such as file management, program execution, and system configuration can be performed quickly with a few typed commands. It also consumes fewer system resources, making it suitable for low performance machines or servers. CLIs provide powerful control over system operations, allowing advanced users and administrators to perform complex configurations. Another advantage is automation, since scripts can execute multiple commands automatically without manual interaction. CLIs are also ideal for remote system management because they work well over network connections.

  Limitations
  A CLI can be difficult for beginners because it requires memorizing commands and understanding syntax rules. There is no visual guidance such as icons or menus, which makes it less intuitive compared to

graphical interfaces. Typing errors can easily lead to incorrect results or system issues. In addition, it is not suitable for tasks that require heavy graphical interaction, such as image editing or design work. Because of these limitations, many modern systems combine CLI and graphical interfaces to provide flexibility and ease of use.

- **Graphical User Interface (GUI)**
  A Graphical User Interface, commonly known as GUI, is a type of user interface that allows users to interact with a computer system through visual elements such as windows, icons, menus, buttons, and pointers. Instead of typing commands, users perform actions by clicking, dragging, and selecting items on the screen. Examples of operating systems that use GUI include Windows 11, macOS, and Ubuntu, a popular distribution of Linux.

  Characteristics
  A GUI is visually oriented and user friendly, using graphical elements to represent files, folders, and system functions. It follows the WIMP model, which stands for windows, icons, menus, and pointer. Users interact with the system using devices such as a mouse, touchpad, or touchscreen. GUIs provide immediate visual feedback, making it easier to understand actions and system responses. They also support multitasking by allowing multiple windows to be open and managed at the same time. Compared to text based interfaces, GUIs require more system resources such as memory and processing power.

  Advantages
  One of the main advantages of a GUI is ease of use, especially for beginners. Users do not need to memorize commands because options are displayed visually. It reduces the chances of errors since users can select from menus instead of typing precise syntax. GUIs are ideal for tasks that involve graphics, multimedia, and design because they allow direct visual manipulation of objects. They also improve productivity for general users by making navigation and file management more intuitive.

  Limitations
  A GUI consumes more system resources compared to a command line interface, which may reduce performance on low specification systems. It

can be slower for advanced users who can execute tasks faster through typed commands. Some complex system configurations may not be easily accessible through graphical menus and may still require command line tools. In addition, GUIs depend heavily on graphical components, so if the graphical environment fails, system access may become limited.

- **Importance of a Good User Interface**

Improved Usability
A well designed interface makes systems easier to learn and operate. Clear navigation, consistent layout, and understandable commands reduce confusion and help users complete tasks efficiently. When users can easily understand how to interact with a system, productivity increases and errors decrease. Good usability also shortens the learning curve, especially for beginners.

Enhanced Accessibility
A good user interface ensures that systems are usable by a wide range of people, including those with disabilities. Features such as readable fonts, proper color contrast, screen reader compatibility, and keyboard navigation support make technology more inclusive. Accessibility improves user satisfaction and ensures that more individuals can effectively use the system.

Better System Control
An effective interface provides users with clear feedback and control over system operations. It shows system status, confirms user actions, and prevents accidental errors through warnings or confirmation messages. When users feel in control and understand what the system is doing, they are more confident and efficient in performing tasks.

Increased Productivity
A good interface streamlines workflows and reduces the number of steps required to accomplish tasks. Logical arrangement of tools, shortcuts, and automation features allow users to work faster. In professional environments, this leads to improved performance and time savings.

Reduced Errors and Frustrations

Poorly designed interfaces can cause mistakes, delays, and frustration. A good user interface minimizes these issues by guiding users clearly and providing helpful feedback. Error prevention mechanisms, such as input validation and clear instructions, contribute to smoother system operation.

— Computer Networks

Computer networks involve connecting computers and devices to share data, resources, and services efficiently and securely. A network is made up of interconnected devices such as computers, servers, routers, smartphones, and IoT devices that exchange information and collaborate. These networks enable Internet access, file sharing, communication through email or video calls, access to cloud services, and the operation of smart systems like IoT devices and automated technologies.

a. *Data Transmission and Protocols*

- ***What is Data Transmission?***
  - Data transmission refers to the process of sending digital information from one device to another across a network.
  - Data transmission sends digital or analog data over a communication medium from one device to another.

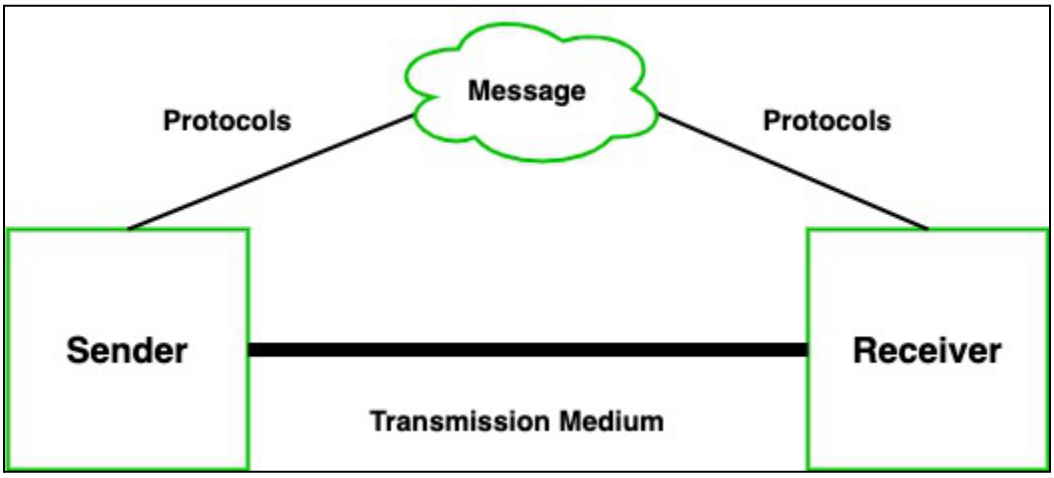
- ***Key Elements of Data Transmission***

☐ **Sender**
- A device that transmits data.
- To transfer a message from the source to the destination, someone must be there who will play the role of the source. The sender plays a role in the data communication system. It is simply a device that sends a data message. The device could be in the form of a computer, mobile, telephone, laptop, video camera, or workstation, etc.

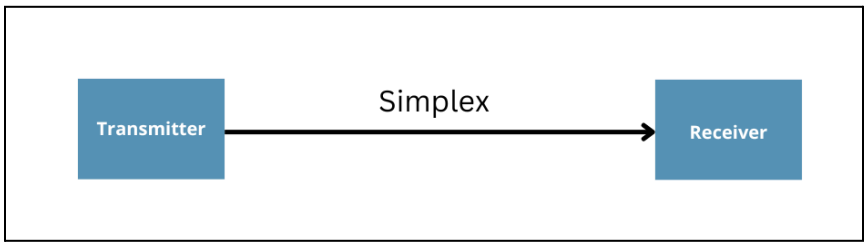
☐ **Receiver**
- A device that receives data.
- It is the destination where finally message sent by the source has arrived. It is a device that receives messages. Same as the sender, the receiver can also be in the form of a computer, a mobile, a workstation, etc.

☐ **Medium**
- path used for transmission
- Acts as a bridge between the sender and receiver. The transmission medium plays that part. It is the path by which data or a message travels from the sender to the receiver. Transmission medium could be guided (with wires) or unguided (without wires), for example, twisted pair cable, fiber optic cable, radio waves, microwaves, etc.

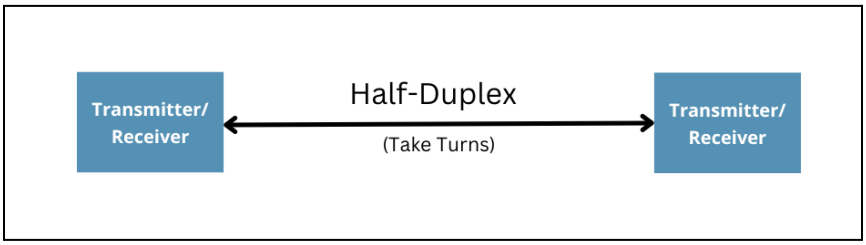
- **Wired Transmission**
  - Also known as the “Guided Media”.
  - Refers to transmission media in which data signals are transmitted through a physical path using cables. The signal is confined and guided along a fixed route, providing controlled communication between network devices.
  - Ex. Ethernet cables, fiber optics
- **Wireless Transmission**
  - Also known as the “Unguided Media”.
  - Uses electromagnetic waves to transmit data without any physical medium. Signals propagate through free space, such as air or vacuum. The main types of unguided media are radio waves, microwaves, and infrared waves.
  - Ex. radio waves, satellites
- **Data Signals**
  - An electrical or electromagnetic signal.
  - A data signal is a form of transmission that conveys information between devices or locations. These signals can be digital or analog, with digital signals standing for data in binary form (0s and 1s), which is particularly common in modern communication systems. Understanding data signals is essential for grasping how devices communicate and share information.
- **Protocols**
  - rules that control communication
  - A set of rules that govern data communication. If two different devices are connected but there is no protocol among them, there would not be any kind of communication between those two devices. Thus, the protocol is necessary for data communication to take place.



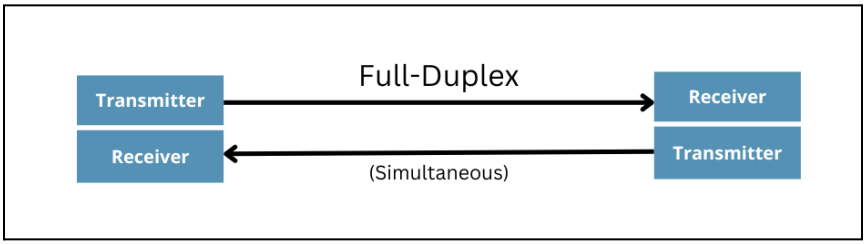
- ***Different Types of Data Transmission***
  - ☐ **Directional Modes:**
    - ❖ **Simplex**
      - Data flows in one direction only, making it ideal for broadcasting applications.
      - In a simplex communication system, the sender can only transmit and the receiver can only receive — there is no mechanism for the receiving end to send information back. This unidirectional nature makes simplex communication straightforward and efficient for scenarios where feedback is unnecessary. Common real-world examples include traditional television and radio broadcasting, where a station transmits signals to millions of passive receivers, as well as keyboard-to-computer input and sensor data feeds in industrial monitoring systems.



    - ❖ **Half-duplex**
      - Data flows in both directions, but only one direction at a time, suitable for error detection and controlled communication.
      - In a half-duplex system, both parties can send and receive information, but must take turns doing so — transmission in one direction must be completed before the other direction can begin. This is similar to a walkie-talkie conversation, where one person speaks while the other listens, and then they switch roles. Half-duplex communication is widely used in CB radio systems, two-way radios, and older Ethernet network configurations.



- **Full-duplex**
  - Data flows simultaneously in both directions, perfect for real-time interactions.
  - Unlike half-duplex systems, full-duplex communication allows both parties to send and receive data at the same time without waiting for the other to finish. This is the mode of operation used in everyday telephone calls, modern mobile networks, video conferencing platforms, and most contemporary internet communications. Full-duplex systems typically achieve bidirectional simultaneous transmission by using separate channels or frequencies for each direction of data flow, preventing the signals from interfering with one another.



- **Synchronization Methods:**
  - **Synchronous**
    - Utilizes clock signals for precise timing, beneficial for high-speed data transfers.
    - In a synchronous system, the sender and receiver share a common clock signal that coordinates exactly when data bits are transmitted and sampled. This tight synchronization eliminates the need for start and stop bits between data packets, allowing for a continuous and highly efficient stream of data. Synchronous communication is commonly used in high-performance applications such as processor-to-memory data transfers, HDMI and DisplayPort video connections, and synchronous DRAM (SDRAM).

- **Asynchronous**
  - Employs start/stop bits, adaptable for varying hardware capabilities.
  - Rather than relying on synchronized timing between sender and receiver, each data packet is wrapped with a start bit to signal the beginning of transmission and one or more stop bits to signal its end. This self-contained framing allows devices with different speeds and capabilities to communicate without needing to be precisely synchronized.

☐ **Bit Transfer Techniques:**

- **Serial**
  - Sends data bits sequentially, optimal for long-distance transfers.
  - Because only a single communication line is required, serial transmission is far less susceptible to the signal degradation and timing inconsistencies that can affect multi-wire systems over longer distances. This simplicity also reduces cabling costs and connector complexity, making serial communication the preferred choice for interfaces such as USB, SATA, PCIe, Ethernet, and fiber optic networks.
- **Parallel**
  - Transmits multiple bits simultaneously, ideal for short-distance, high-volume data movement.
  - By sending several bits at once — typically 8, 16, 32, or 64 bits in a single clock cycle — parallel interfaces can theoretically achieve much higher raw data transfer rates than serial connections at the same clock speed. Parallel communication has historically been used in applications such as internal processor buses, older printer ports (IEEE 1284), and legacy IDE hard drive interfaces.

- ***Packet Switching***

  Most networks transmit data using packet switching.

☐ **What is packet switching?**

Packet switching is a method for moving data across a packet-switching network. Instead of sending your data in one big piece, it breaks it down into smaller chunks called "packets." Each of these packets can travel independently through the network, taking different paths to reach the same destination

☐ **What is a Packet?**

A packet is a small unit of data used in network communication, particularly within the TCP/IP model, which governs internet data transfer. These IP packets contain both data and control information that helps them navigate the network efficiently.

☐ **Structure of a Packet:**

1) **Header:**
   The header contains instructions about the data carried by the packet. These instructions may include:
   - Length of packet (some networks have fixed-length packets, while others rely on the header to contain this information)
   - Synchronization (a few bits that help the packet match up to the network)
   - Packet number (which packet this is in a sequence of packets)
   - Protocol (on networks that carry multiple types of information, the protocol defines what type of packet is being transmitted: e-mail, Web page, streaming video)
   - Destination address (where the packet is going)
   - Originating address (where the packet came from)
2) **Payload (Data):**
   Also called the body or data of a packet. This is the actual data that the packet is delivering to the destination. If a packet is fixed-length, then the payload may be padded with blank information to make it the right size.
3) **Footer:**
   The trailer, sometimes called the footer, typically contains a couple of bits that tell the receiving device that it has reached the end of the packet. It may also include error checking. The most common error checking used in packets is the Cyclic Redundancy Check (CRC).

- ***Protocols in Networking***

☐ **What are protocols?**

- A protocol is a set of rules that ensures devices communicate correctly.
- Protocols define:

  Data Format
  - Protocols define the structure or format of data so that devices can correctly interpret what they receive. This includes organizing the data into parts such as headers, payloads, and trailers.

  Transmission Speed
  - Protocols also influence how fast data is transmitted across a network. They may define the data rate or implement flow control mechanisms to prevent overwhelming the receiver. For instance, Ethernet standards specify speeds such as 100 Mbps or 1 Gbps, while TCP dynamically adjusts transmission speed based on network conditions. This ensures efficient communication without congestion or data loss.

Error Handling

- Error handling is another important function defined by protocols. During transmission, data can become corrupted due to noise or interference, so protocols include mechanisms like checksums or cyclic redundancy checks (CRC) to detect errors. If an error is detected, protocols such as TCP request retransmission of the damaged data. This ensures that the information received is accurate and complete.

Acknowledgment

- Protocols determine whether the receiver must confirm successful data delivery. In reliable communication protocols like TCP, the receiver sends an acknowledgment (ACK) to inform the sender that the data was received correctly. If no acknowledgment is received within a certain time, the sender retransmits the data. In contrast, User Datagram Protocol (UDP) does not require acknowledgments, making it faster but less reliable.

Routing

- Protocols also define how data travels from the source to the destination across networks. The Internet Protocol (IP) is responsible for logical addressing and routing packets to their destination. Additionally, routing protocols such as Open Shortest Path First (OSPF) help routers determine the best available path for data transmission. This ensures that packets reach the correct destination efficiently.

- ***Important Protocols***

Computer networks rely on various protocols to enable devices to communicate efficiently and reliably. These protocols act as standardized rules that govern how data is transmitted, received, addressed, and processed across networks.

Here are some key protocols that are widely used in computer networks:

☐ **TCP (Transmission Control Protocol):**

Ensures data is sent and received accurately by breaking it into packets, sending them, and reassembling them at the destination.

- Used in web browsing to ensure the complete and accurate delivery of web page data.
- Used in file transfers to guarantee that files are received without corruption.
- Used in email communication to reliably send and receive messages.

- ☐ **IP (Internet Protocol):**

  Addresses and routes the packets to make sure they reach the right destination.

  - Used to assign logical addresses (IP addresses) to devices on a network.
  - Used to route packets from source to destination across multiple networks.
  - Used to enable communication between devices on local and wide area networks (LAN/WAN).

- ☐ **HTTP/HTTPS (HyperText Transfer Protocol/Secure):**

  HTTP is used for transferring web pages on the Internet. When you browse a website, your browser uses HTTP to request and display web pages. HTTPS is a secure version of HTTP that encrypts data to protect it from being intercepted.

  - Used for loading websites in web browsers.
  - Used for transferring web resources such as images, videos, and scripts.
  - HTTPS is used for secure online transactions, such as online banking and e-commerce.

- ☐ **FTP (File Transfer Protocol):**

  Used for transferring files between computers on a network. It allows users to upload and download files.SMTP (Simple Mail Transfer Protocol): Used for sending emails. It transfers emails from a client to a server or between servers.

  - Used to upload website files to a web server.
  - Used to download large files from remote servers.
  - Used for backup and file sharing between computers in a network.

- ☐ **DNS (Domain Name System):**

  It is used to translate human-readable domain names (like www.example.com) into IP addresses that computers use to identify each other on the network.

  - Used to translate domain names into IP addresses.
  - Used to help browsers locate website servers.
  - Used in email delivery to identify mail servers through DNS records.

- ☐ **DHCP (Dynamic Host Configuration Protocol):**

  Automatically assigns IP addresses to devices on a network, ensuring each device has a unique address.

  - Used to automatically assign IP addresses to devices.

- Used to provide default gateway and DNS server information.
- Used in large networks to manage IP address allocation efficiently.

☐ **UDP (User Datagram Protocol):**

User Datagram Protocol (UDP) is a Transport Layer protocol of the Internet Protocol (IP) that provides fast, connectionless, and lightweight communication between processes. It does not guarantee delivery, order, or error checking, making it suitable for real-time and time-sensitive applications such as video streaming, DNS, and VoIP.

- Used in video streaming services where speed is more important than reliability.
- Used in online gaming for fast real-time communication.
- Used in VoIP (Voice over Internet Protocol) calls to transmit voice data quickly.

b. *Network Security and Management* ——————————————————

Networks must be protected from unauthorized access and attacks. Networks are vulnerable to threats, so protecting them is essential.

- ***Network Security***

  What is **"Network Security"?**

  - Network security refers to protecting network systems and data from unauthorized access, misuse, or attacks.
  - Network security is the strategic combination of hardware and software designed to protect sensitive data in a computer network. Network access controls, intrusion detection, and many other types of network security functions work together to secure the environment against unauthorized access, data breaches, malware delivery, and other cyberattacks.

  ☐ **Common Network Threats**

  - Network security threats refer to any potential danger to network integrity, confidentiality, or availability. These threats manifest as unauthorized access attempts, data interception, service disruptions, or infrastructure compromises.

    Common network threats include:

    ❖ **Hacking**
    - Unauthorized access
    - Hacking in cybersecurity refers to the misuse of devices like computers, smartphones, tablets, and networks to cause damage to or corrupt systems, gather information on users, steal data and documents, or disrupt data-related activity.

    ❖ **Malware**
    - Malware is a program that attacks information systems. There are various types of malware, each designed to perform specific malicious activities. For example, ransomware encrypts files and holds them for ransom, spyware covertly spies on victims, and Trojans infiltrate systems.

    ❖ **Phishing**
    - Phishing is a type of fraud that occurs when a threat actor impersonates a reputable entity in person, via email, or other communication forms. Threat actors often use phishing emails to spread malicious attachments or links that perform various functions, such as extracting the victim's account information or login credentials.

- **DoS/DDoS Attacks**
  - A denial-of-service (DoS) attack floods a server with traffic, making a website or resource unavailable. A distributed denial-of-service (DDoS) attack is a DoS attack that uses multiple computers or machines to flood a targeted resource. Both types of attacks overload a server or web application with the goal of interrupting services.
  - The principal difference between a DoS attack and a DDoS attack is that the former is a system-on-system attack, while the latter involves several systems attacking a single system.

☐ **Security Mechanisms**

- Security mechanisms are technical tools and techniques that are used to implement security services. A mechanism might operate by itself or with others to provide a particular service.
- Security mechanisms are techniques and controls used to protect data and systems from unauthorized access, attacks, and misuse by ensuring confidentiality, integrity, and availability.

- **Firewalls**
  - Acts as a security gate.
  - A firewall is a network security device that separates a trusted internal network from an external network deemed untrustworthy, such as the Internet. It regulates incoming and outgoing network traffic based on preset security rules. Firewalls are paramount in shielding networks from unauthorized access, harmful activities, and potential threats, and can exist as hardware, software, software-as-a-service (SaaS), or public or private (virtual) cloud.
- **Encryption**
  - Conversion of data into an unreadable form.
  - Encryption is a way of scrambling data so that only authorized parties can understand the information. In technical terms, it is the process of converting human-readable plaintext to incomprehensible text, also known as ciphertext. In simpler terms, encryption takes readable data and alters it so that it appears random. Encryption requires the use of a cryptographic key: a set of mathematical values that both the sender and the recipient of an encrypted message agree on.
- **Authentication**
  - Verifies identity

- Authentication is the process of determining whether someone or something is who or what they say they are. Authentication technology provides access control for systems by checking to see if a user's credentials match the credentials in a database of authorized users or a data authentication server. In doing this, authentication ensures that systems, processes, and enterprise information are secure.

❖ **Intrusive Detection Systems (IDS)**
- IDS monitors the network activity for suspicious behavior.
- An intrusion detection system (IDS) is an application that monitors network traffic and searches for known threats and suspicious or malicious activity. The IDS sends alerts to IT and security teams when it detects any security risks and threats.

❖ **Access Control**
- Controls who can use the network
- Access control is a data security process that enables organizations to manage who is authorized to access corporate data and resources. Secure access control uses policies that verify users are who they claim to be and ensure appropriate control access levels are granted to users.

Aside from security tools and mechanisms like firewalls, encryption, authentication, IDS, and access control, network security is also guided by a framework known as the CIA Triad.

The CIA Triad
- The three letters in "CIA triad" stand for Confidentiality, Integrity, and Availability. The CIA triad is a common model that forms the basis for the development of security systems. They are used for finding vulnerabilities and methods for creating solutions.
- The confidentiality, integrity, and availability of information is crucial to the operation of a business, and the CIA triad segments these three ideas into separate focal points. This differentiation is helpful because it helps guide security teams as they pinpoint the different ways in which they can address each concern.

- ***Network Management***
  What is **"Network Management"?**
  - Network management ensures smooth and efficient network operation.
  - Network management refers to two related concepts. First is the process of configuring, monitoring, and managing the performance of a network. Second is the platform that IT and NetOps teams use to complete these ongoing tasks.

☐ **Main Network Management Tasks** - Main Network Management Tasks are the essential activities involved in managing and maintaining a computer network to ensure it runs smoothly, efficiently, and securely. These tasks include monitoring network performance, detecting and resolving problems, configuring and updating devices, tracking resource usage, and protecting the network from security threats.

❖ **Monitoring traffic** - Network traffic analysis (NTA) is a technique used by network administrators to examine network activity, manage availability, and identify unusual activity.

❖ **Managing network devices (routers, switches)** - Network devices such as routers and switches are the backbone of any network. Managing these devices includes configuring them properly, maintaining their operational status, and ensuring they communicate effectively.

- **Optimizing performance** - Network performance can be affected by heavy traffic, inefficient routing, or outdated hardware. Optimizing performance involves analyzing network usage, balancing loads, prioritizing critical data, and making adjustments to improve speed, reliability, and overall efficiency. This ensures users experience minimal delays or interruptions.
- **Troubleshooting faults** - Despite careful management, networks can experience problems such as hardware failures, software glitches, or connectivity issues.
- **Updating firmware and configurations** - Keeping network devices up to date is essential for security and functionality. This task includes installing firmware updates, patching vulnerabilities, and adjusting device configurations to improve performance, add new features, and prevent potential exploits. Regular updates ensure the network remains robust and protected against evolving threats.

☐ **Network Monitoring Tools**

- Network management ensures smooth and efficient network operation.
- Network monitoring tools are software or hardware solutions used to observe, analyze, and manage network traffic and device performance. They help administrators detect issues, prevent downtime, and optimize the network's efficiency.
  - **SNMP (Simple Network Management Protocol)** – A protocol used to collect and organize information from network devices like routers, switches, and servers. It allows administrators to monitor device status, track performance metrics, and receive alerts for faults or unusual activity.
  - **Wireshark (packet analysis)** – A powerful tool that captures and inspects network packets in real-time. It helps administrators analyze network traffic, troubleshoot issues, detect intrusions, and understand how data flows across the network.
  - **Network performance dashboards** – Visual interfaces that display real-time data on network health, traffic, and device performance. Dashboards allow administrators to quickly spot bottlenecks, monitor bandwidth usage, and identify potential problems before they impact users.

c. *Internet and Wireless Communication* ————————————————

- Modern networks include both wired and wireless technologies.

- ***Internet Technologies***

  What is **"Internet Technologies"?** - That group of technologies that allow users to access information and communication over the World Wide Web (Web browsers, ftp, e-mail, associated hardware, Internet service providers, and so forth).

  ☐ **How the Internet Works** - The Internet works by connecting devices like computers and smartphones to each other through Internet Service Providers (ISPs), which act as gateways to the larger network. When data is sent from one device to another, it travels through a series of routers and high-capacity backbone networks that direct it along the most efficient path. Communication over the Internet relies on the TCP/IP protocol, which breaks information into packets, ensures they are delivered correctly, and reassembles them at the destination, allowing seamless and reliable exchange of data across the globe.

  ☐ **Key Internet Services**

  ❖ **Web Browsing**

  - Allows users to access and navigate websites on the Internet using browsers like Chrome or Firefox.
  - It serves as the primary gateway through which people consume information, shop online, access services, and interact with web-based applications. Modern browsers do far more than simply display text and images — they run complex web applications, stream high-definition video, support secure financial transactions, and render interactive 3D content. Features such as tabbed browsing, bookmark management, private/incognito modes, and built-in password managers have made browsers an essential productivity tool for both personal and professional use.

  ❖ **Cloud Computing**

  - It allows individuals and businesses to store data, run software, and manage workloads without relying on local hardware, making services scalable and accessible from anywhere.
  - Rather than investing in expensive on-site servers and infrastructure, organizations can rent computing resources from providers such as Amazon Web Services (AWS),

Microsoft Azure, or Google Cloud on a pay-as-you-go basis. This model offers significant cost savings, as businesses only pay for the resources they actually use. Cloud computing also enables rapid scalability — companies can instantly increase or decrease their computing capacity in response to changing demand, without the delays associated with purchasing and installing physical hardware. From data backup and disaster recovery to artificial intelligence workloads and global application deployment, cloud computing has become the backbone of the modern digital economy.

- **Email**
  - A digital communication service that lets users send and receive messages, documents, and multimedia files instantly.
  - Since its introduction, it has become one of the most widely used communication tools in both personal and professional settings, with billions of messages exchanged every day. Modern email platforms such as Gmail, Outlook, and Yahoo Mail offer far more than basic messaging — they include calendar integration, task management, spam filtering, automated sorting, and cloud storage attachments. In business environments, email serves as the official record of communications, contract negotiations, and project updates. Security features such as end-to-end encryption, two-factor authentication, and phishing detection have been developed to protect users from the growing threat of email-based cyberattacks, including phishing scams and malware-laden attachments.
- **Social Media**
  - Social media platforms enable users to create, share, and interact with content and connect with others globally.
  - Platforms such as Facebook, Instagram, X (formerly Twitter), LinkedIn, TikTok, and YouTube have fundamentally changed how people communicate, consume news, build communities, and conduct business. For individuals, social media provides a space to share personal milestones, follow interests, and maintain relationships across distances. For

businesses, it has become a powerful marketing and customer engagement tool, enabling brands to reach targeted audiences, run advertising campaigns, and receive real-time feedback from consumers. However, social media also raises significant concerns around data privacy, misinformation, cyberbullying, and its psychological impact on users, particularly younger audiences.

❖ **Remote Work Systems**

- Remote work systems provide tools and platforms for employees to work outside traditional offices, often from home.
- hese systems encompass a broad range of technologies including video conferencing platforms like Zoom and Microsoft Teams, project management tools like Asana and Trello, cloud-based document collaboration through Google Workspace or Microsoft 365, and secure network access via Virtual Private Networks (VPNs). The widespread adoption of remote work accelerated dramatically during the COVID-19 pandemic and has since reshaped expectations around workplace flexibility for both employers and employees. Organizations benefit from access to a wider global talent pool and reduced overhead costs, while employees gain greater autonomy and improved work-life balance.

☐ **Cloud Computing and Networks**

- Networks enable cloud services like Google Drive, AWS, and Microsoft Azure by allowing data and applications to be stored and accessed remotely over the Internet. These networks connect users' devices to powerful servers, ensuring that files, software, and computing resources can be managed, shared, and used from anywhere, providing flexibility, scalability, and real-time collaboration without relying on local storage or hardware.

- ***Wireless Communication***

What is **"Wireless Communication"?** - Wireless Communication is a method of transmitting data or information between two or more devices without the use of physical wires or cables. Instead of using cables, it uses radio waves (RF - Radio

Frequency) or other types of electromagnetic waves to send signals through the air or free space.

- ☐ **Major Wireless Technologies**
  - ❖ **Wi-FI**
    - A wireless LAN technology that allows devices to connect to the Internet and local networks without cables. It is commonly used in homes, schools, and offices, and operates based on the IEEE 802.11 standards to provide reliable, high-speed wireless connectivity.
  - ❖ **Bluetooth**
    - Bluetooth is a short-range wireless communication technology used to connect devices like headphones, keyboards, and IoT gadgets. It is designed for low power consumption, making it ideal for small, battery-powered devices that need to communicate over short distances.
  - ❖ **Cellular Networks (3G, 4G, 5G)**
    - Cellular networks provide mobile connectivity over wide areas, allowing users to access voice and data services on the go. 4G networks support high-speed internet for streaming and downloads, while 5G offers ultra-fast speeds, extremely low latency, and the ability to connect a massive number of IoT devices simultaneously.
  - ❖ **Satellite Communication**
    - Satellite communication is used to provide connectivity in remote areas where traditional networks are unavailable. It supports services like GPS navigation and global broadcasting, ensuring communication and data access on a worldwide scale.
- ☐ **Wireless Network Challenges**
  - ❖ **Signal Interference**
    - Signal interference occurs when other electronic devices or networks disrupt wireless communication, causing slower speeds or dropped connections.
    - Common sources include microwave ovens, Bluetooth devices, baby monitors, and neighboring Wi-Fi networks operating on the same frequency channel. In densely populated environments like apartment buildings or open-plan offices, interference becomes especially severe

as dozens of networks compete for limited radio spectrum. The result is increased packet loss, higher latency, and unpredictable connection quality that frustrates both users and administrators.

- **Security Risks**
  - Wireless networks are vulnerable to unauthorized access, eavesdropping, and hacking if not properly secured.
  - Because wireless signals travel through the air rather than through physical cables, any device within range can attempt to intercept transmitted data. Common attack methods include Man-in-the-Middle (MitM) attacks, rogue access points designed to impersonate legitimate networks, and brute-force attempts to crack weak passwords. Without strong security protocols such as WPA3 encryption and multi-factor authentication, sensitive data including passwords, financial information, and private communications remains exposed to malicious actors.
- **Limited Bandwidth**
  - Bandwidth limits the amount of data that can be transmitted at a time, affecting network speed and performance.
  - As more devices connect to a wireless network simultaneously, available bandwidth is divided among all users, causing noticeable slowdowns during peak usage periods. Bandwidth-heavy activities such as video streaming, video conferencing, and large file transfers can quickly saturate a network, leaving other users with degraded service. Unlike wired connections that can be physically upgraded with higher-capacity cables, wireless bandwidth is constrained by the capabilities of the router hardware and the wireless standard in use.
- **Range Limitations**
  - Wireless signals weaken over distance or through obstacles, reducing coverage and connectivity quality.
  - Physical barriers such as thick concrete walls, metal structures, floors between building levels, and even large furniture can significantly attenuate a wireless signal. The further a device is from the access point, the weaker the signal strength and the lower the achievable data rate. In

large buildings or outdoor environments, dead zones — areas with little to no wireless coverage — are a frequent problem that can disrupt productivity and user experience.

- **Strong Encryption**
  - Encryption protects data from unauthorized access, keeping wireless communications secure.
  - Modern encryption standards such as WPA3 scramble transmitted data into an unreadable format that can only be decoded by authorized recipients with the correct decryption key. This ensures that even if a signal is intercepted, the data remains protected and unusable to attackers. Organizations should enforce end-to-end encryption across all wireless communications and regularly update encryption protocols to guard against newly discovered vulnerabilities, ensuring sensitive information stays confidential at all times.
- **Better Antennas**
  - Improved antennas enhance signal strength, coverage, and reliability.
  - High-gain directional antennas can focus wireless signals toward specific areas, extending range and reducing interference from unintended directions. Omnidirectional antennas, meanwhile, provide broader 360-degree coverage suitable for open environments. Technologies such as MIMO (Multiple Input, Multiple Output) use multiple antennas simultaneously to transmit and receive data across several spatial streams, dramatically increasing throughput and connection stability. Upgrading antenna hardware is one of the most cost-effective ways to improve overall wireless network performance.
- **Frequency Management**
  - Managing frequencies reduces interference and optimizes network performance.
  - Modern wireless networks operate across multiple frequency bands, primarily 2.4 GHz and 5 GHz, each with its own set of channels. Careful channel planning — ensuring neighboring access points operate on non-overlapping channels — minimizes co-channel interference and improves overall network efficiency. Advanced spectrum analysis tools allow

administrators to monitor frequency usage in real time, identify congested channels, and automatically reassign traffic to cleaner frequencies. Dual-band and tri-band routers further help by intelligently distributing devices across available bands based on their needs.

❖ **Advanced Protocols**

- Advanced protocols improve efficiency, security, and stability in wireless communication.
- Standards such as Wi-Fi 6 (802.11ax) and the emerging Wi-Fi 7 introduce technologies like OFDMA (Orthogonal Frequency Division Multiple Access), which allows access points to serve multiple devices simultaneously rather than sequentially, dramatically reducing latency in crowded environments. Target Wake Time (TWT) technology helps battery-powered devices conserve energy by scheduling precise times to send and receive data. These newer protocols also incorporate stronger built-in security features and more efficient data compression, making wireless networks faster, more reliable, and better suited to the growing demands of modern connected devices.

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